GARP and Afriat's Theorem
Production

Econ 2100 Fall 2018

Lecture 8, September 24

Outline

1. Generalized Axiom of Revealed Preferences
2. Afriat's Theorem
3. Production Sets and Production Functions
4. Profits Maximization, Supply Correspondence, and Profit Function
5. Hotelling and Shephard Lemmas
6. Cost Minimization

From Last Class

- $\{x^j, p^j, w^j\}_{j=1}^N$ is a finite set of demand data that satisfy $p^j \cdot x^j \leq w^j$.
- Directly revealed weak preference: $x^j \succsim^R x^k$ if $p^j \cdot x^k \leq w^j$
- Directly revealed strict preference: $x^j \succ^R x^k$ if $p^j \cdot x^k < w^j$
- Indirectly revealed weak preference:

$$x^j \succsim^I x^k \text{ if } \exists\ x^{i_1}, \ldots, x^{i_m} \text{ s.t. } x^j \succsim^R x^{i_1}, ..., x^{i_m} \succsim^R x^k$$

- Indirectly revealed strict preference:

$$x^j \succ^I x^k \text{ if } \exists\ x^{i_1}, \ldots, x^{i_m} \text{ s.t. } x^j \succsim^R x^{i_1}, ..., x^{i_m} \succsim^R x^k \text{ with one strict}$$

- Using these definitions, we can find a condition that guarantees choices are the result of the maximizing a preference relation or a utility function.

Generalized Axiom of Revealed Preference

Axiom (Generalized Axiom of Revealed Preference - GARP)

If $x^j \succsim^R x^k$, then $\text{not}\left[x^k \succ^I x^j\right]$.

Generalized Axiom of Revealed Preference

Axiom (Generalized Axiom of Revealed Preference - GARP)

If $x^j \succsim^R x^k$, then $\text{not}\left[x^k \succ^I x^j\right]$.

- If a choice is directly revealed weakly prefered to a bundle, this bundle cannot be indirectly revealed strictly preferred to that choice.

Generalized Axiom of Revealed Preference

Axiom (Generalized Axiom of Revealed Preference - GARP)

If $x^j \succsim^R x^k$, then $\text{not}\left[x^k \succ^I x^j\right]$.

- If a choice is directly revealed weakly prefered to a bundle, this bundle cannot be indirectly revealed strictly preferred to that choice.
- Note that if $p^j \cdot x^j < w^j$ then $x^j \succ^R x^j$ and therefore $x^j \succ^I x^j$ and therefore GARP cannot hold.

Generalized Axiom of Revealed Preference

Axiom (Generalized Axiom of Revealed Preference - GARP)

If $x^j \succsim^R x^k$, then $\text{not}\left[x^k \succ^I x^j\right]$.

- If a choice is directly revealed weakly prefered to a bundle, this bundle cannot be indirectly revealed strictly preferred to that choice.
- Note that if $p^j \cdot x^j < w^j$ then $x^j \succ^R x^j$ and therefore $x^j \succ^I x^j$ and therefore GARP cannot hold.
  - Therefore, GARP implies the consumer spends all her money.

Generalized Axiom of Revealed Preference

Axiom (Generalized Axiom of Revealed Preference - GARP)

If $x^j \succsim^R x^k$, then $\text{not}\left[x^k \succ^I x^j\right]$.

- If a choice is directly revealed weakly prefered to a bundle, this bundle cannot be indirectly revealed strictly preferred to that choice.
- Note that if $p^j \cdot x^j < w^j$ then $x^j \succ^R x^j$ and therefore $x^j \succ^I x^j$ and therefore GARP cannot hold.
  - Therefore, GARP implies the consumer spends all her money.
- Now suppose $x^j \succsim^R x^k$ and $x^k \succ^I x^j$. Then there is a chain of weak preference from j to k, and a chain back (from k to j) that has at least one strict preference.

Generalized Axiom of Revealed Preference

Axiom (Generalized Axiom of Revealed Preference - GARP)

If $x^j \succsim^R x^k$, then $\text{not}\left[x^k \succ^I x^j\right]$.

- If a choice is directly revealed weakly prefered to a bundle, this bundle cannot be indirectly revealed strictly preferred to that choice.
- Note that if $p^j \cdot x^j < w^j$ then $x^j \succ^R x^j$ and therefore $x^j \succ^I x^j$ and therefore GARP cannot hold.
  - Therefore, GARP implies the consumer spends all her money.
- Now suppose $x^j \succsim^R x^k$ and $x^k \succ^I x^j$. Then there is a chain of weak preference from j to k, and a chain back (from k to j) that has at least one strict preference.
  - This is a cycle with a strict preference inside; if such a cycle exists one can construct a "money pump" to make money off the consumer and leave her exactly at the bundle she started from.

Generalized Axiom of Revealed Preference

Axiom (Generalized Axiom of Revealed Preference - GARP)

If $x^j \succsim^R x^k$, then $\text{not}\left[x^k \succ^I x^j\right]$.

- If a choice is directly revealed weakly prefered to a bundle, this bundle cannot be indirectly revealed strictly preferred to that choice.
- Note that if $p^j \cdot x^j < w^j$ then $x^j \succ^R x^j$ and therefore $x^j \succ^I x^j$ and therefore GARP cannot hold.
  - Therefore, GARP implies the consumer spends all her money.
- Now suppose $x^j \succsim^R x^k$ and $x^k \succ^I x^j$. Then there is a chain of weak preference from j to k, and a chain back (from k to j) that has at least one strict preference.
  - This is a cycle with a strict preference inside; if such a cycle exists one can construct a "money pump" to make money off the consumer and leave her exactly at the bundle she started from.
  - GARP rules these cycles out.

Generalized Axiom of Revealed Preference

Axiom (Generalized Axiom of Revealed Preference - GARP)

If $x^j \succsim^R x^k$, then $\text{not}\left[x^k \succ^I x^j\right]$.

- If a choice is directly revealed weakly prefered to a bundle, this bundle cannot be indirectly revealed strictly preferred to that choice.
- Note that if $p^j \cdot x^j < w^j$ then $x^j \succ^R x^j$ and therefore $x^j \succ^I x^j$ and therefore GARP cannot hold.
  - Therefore, GARP implies the consumer spends all her money.
- Now suppose $x^j \succsim^R x^k$ and $x^k \succ^I x^j$. Then there is a chain of weak preference from j to k, and a chain back (from k to j) that has at least one strict preference.
  - This is a cycle with a strict preference inside; if such a cycle exists one can construct a "money pump" to make money off the consumer and leave her exactly at the bundle she started from.
  - GARP rules these cycles out.

Problem 4, Problem Set 4.

Show that GARP is equivalent to the following: $\boxed{\text{If } x^j \succsim^I x^k \text{ then } \text{not}\left[x^k \succ^I x^j\right]}$.

Utility and Demand Data

Definition

A utility function $u : \mathbf{R}^n_+ \to \mathbf{R}$ rationalizes a finite set of demand data $\{x^j, p^j, w^j\}_{j=1}^N$ if

$$u(x^j) \geq u(x) \qquad \text{whenever} \qquad p^j \cdot x \leq w^j$$

Utility and Demand Data

Definition

A utility function $u : \mathbf{R}^n_+ \to \mathbf{R}$ rationalizes a finite set of demand data $\{x^j, p^j, w^j\}_{j=1}^N$ if

$$u(x^j) \geq u(x) \qquad \text{whenever} \qquad p^j \cdot x \leq w^j$$

- This definition can alternatively be stated by the condition

$$x^j \in \arg\max_{x \in \mathbf{R}^n_+} u(x) \text{ subject to } p^j \cdot x \leq w^j$$

Utility and Demand Data

Definition

A utility function $u : \mathbf{R}^n_+ \to \mathbf{R}$ rationalizes a finite set of demand data $\{x^j, p^j, w^j\}_{j=1}^N$ if

$$u(x^j) \geq u(x) \qquad \text{whenever} \qquad p^j \cdot x \leq w^j$$

- This definition can alternatively be stated by the condition

$$x^j \in \arg\max_{x \in \mathbf{R}^n_+} u(x) \text{ subject to } p^j \cdot x \leq w^j$$

- If the observations satisfy the definition, the data is consistent with the behavior of an individual who, in each observation, chooses a utility-maximizing bundle subject to the corresponding budget constraint.

Afriat's Theorem

Theorem (Afriat)

Given a finite set of demand data $\{(x^j, p^j, w^j)\}_{k=1}^N$, *the following are equivalent:*

Afriat's Theorem

Theorem (Afriat)

Given a finite set of demand data $\{(x^j, p^j, w^j)\}_{k=1}^N$, *the following are equivalent:*

1. *There exists a locally nonsatiated utility function which rationalizes the data.*

Afriat's Theorem

Theorem (Afriat)

Given a finite set of demand data $\{(x^j, p^j, w^j)\}_{k=1}^N$, *the following are equivalent:*

1. *There exists a locally nonsatiated utility function which rationalizes the data.*
2. *The data satisfy the Generalized Axiom of Revealed Preference.*

Afriat's Theorem

Theorem (Afriat)

Given a finite set of demand data $\{(x^j, p^j, w^j)\}_{k=1}^N$, *the following are equivalent:*

1. *There exists a locally nonsatiated utility function which rationalizes the data.*
2. *The data satisfy the Generalized Axiom of Revealed Preference.*
3. *There exist numbers* $v^j, \lambda^j > 0$ *such that*

$$v^k \leq v^j + \lambda^j [p^j \cdot x^k - w^j], \qquad \text{for all } j, k = 1, \ldots, N \tag{G}$$

Afriat's Theorem

Theorem (Afriat)

Given a finite set of demand data $\{(x^j, p^j, w^j)\}_{k=1}^N$, *the following are equivalent:*

1. *There exists a locally nonsatiated utility function which rationalizes the data.*
2. *The data satisfy the Generalized Axiom of Revealed Preference.*
3. *There exist numbers* $v^j, \lambda^j > 0$ *such that*
$$v^k \leq v^j + \lambda^j[p^j \cdot x^k - w^j], \qquad \text{for all } j, k = 1, \ldots, N \tag{G}$$
4. *There exists a continuous, strictly increasing, and concave utility function which rationalizes the data.*

Afriat's Theorem

Theorem (Afriat)

Given a finite set of demand data $\{(x^j, p^j, w^j)\}_{k=1}^N$, *the following are equivalent:*

1. *There exists a locally nonsatiated utility function which rationalizes the data.*
2. *The data satisfy the Generalized Axiom of Revealed Preference.*
3. *There exist numbers* $v^j, \lambda^j > 0$ *such that*
$$v^k \leq v^j + \lambda^j[p^j \cdot x^k - w^j], \qquad \text{for all } j, k = 1, \ldots, N \tag{G}$$
4. *There exists a continuous, strictly increasing, and concave utility function which rationalizes the data.*

- Under GARP, the consumer behaved as if each choice was optimal (i.e. utility-maximizing).

Afriat's Theorem

Theorem (Afriat)

Given a finite set of demand data $\{(x^j, p^j, w^j)\}_{k=1}^N$, *the following are equivalent:*

1. *There exists a locally nonsatiated utility function which rationalizes the data.*
2. *The data satisfy the Generalized Axiom of Revealed Preference.*
3. *There exist numbers* $v^j, \lambda^j > 0$ *such that*
$$v^k \leq v^j + \lambda^j[p^j \cdot x^k - w^j], \qquad \text{for all } j, k = 1, \ldots, N \tag{G}$$
4. *There exists a continuous, strictly increasing, and concave utility function which rationalizes the data.*

- Under GARP, the consumer behaved as if each choice was optimal (i.e. utility-maximizing).
- The fact (4) implies (1) is immediate (strictly increasing implies locally non satiated). We will skip the remainder of the proof, but see Kreps or Varian for it.

Afriat's Theorem

Theorem (Afriat)

Given a finite set of demand data $\{(x^j, p^j, w^j)\}_{k=1}^N$, *the following are equivalent:*

1. *There exists a locally nonsatiated utility function which rationalizes the data.*
2. *The data satisfy the Generalized Axiom of Revealed Preference.*
3. *There exist numbers* $v^j, \lambda^j > 0$ *such that*
$$v^k \leq v^j + \lambda^j[p^j \cdot x^k - w^j], \qquad \text{for all } j, k = 1, \ldots, N \tag{G}$$
4. *There exists a continuous, strictly increasing, and concave utility function which rationalizes the data.*

Afriat's Theorem

Theorem (Afriat)

Given a finite set of demand data $\{(x^j, p^j, w^j)\}_{k=1}^N$, *the following are equivalent:*

1. *There exists a locally nonsatiated utility function which rationalizes the data.*
2. *The data satisfy the Generalized Axiom of Revealed Preference.*
3. *There exist numbers* $v^j, \lambda^j > 0$ *such that*
$$v^k \leq v^j + \lambda^j[p^j \cdot x^k - w^j], \qquad \text{for all } j, k = 1, \ldots, N \tag{G}$$
4. *There exists a continuous, strictly increasing, and concave utility function which rationalizes the data.*

- For finite data, the only implication of preference maximization is GARP; conversely, if the data violate GARP, the consumer is not maximizing any continuous, strictly monotone, and convex preference relation.

Afriat's Theorem

Theorem (Afriat)

Given a finite set of demand data $\{(x^j, p^j, w^j)\}_{k=1}^N$, *the following are equivalent:*

1. *There exists a locally nonsatiated utility function which rationalizes the data.*
2. *The data satisfy the Generalized Axiom of Revealed Preference.*
3. *There exist numbers* $v^j, \lambda^j > 0$ *such that*
$$v^k \leq v^j + \lambda^j[p^j \cdot x^k - w^j], \qquad \text{for all } j, k = 1, \ldots, N \tag{G}$$
4. *There exists a continuous, strictly increasing, and concave utility function which rationalizes the data.*

- For finite data, the only implication of preference maximization is GARP; conversely, if the data violate GARP, the consumer is not maximizing any continuous, strictly monotone, and convex preference relation.
- Using a finite number of observations, one cannot distinguish a continuous, strictly monotone, and convex preference from a locally nonsatiated one.

Afriat's Theorem

Theorem (Afriat)

Given a finite set of demand data $\{(x^j, p^j, w^j)\}_{k=1}^N$, *the following are equivalent:*

1. *There exists a locally nonsatiated utility function which rationalizes the data.*
2. *The data satisfy the Generalized Axiom of Revealed Preference.*
3. *There exist numbers* $v^j, \lambda^j > 0$ *such that*
$$v^k \leq v^j + \lambda^j[p^j \cdot x^k - w^j], \qquad \text{for all } j, k = 1, \ldots, N \tag{G}$$
4. *There exists a continuous, strictly increasing, and concave utility function which rationalizes the data.*

- For finite data, the only implication of preference maximization is GARP; conversely, if the data violate GARP, the consumer is not maximizing any continuous, strictly monotone, and convex preference relation.
- Using a finite number of observations, one cannot distinguish a continuous, strictly monotone, and convex preference from a locally nonsatiated one.
- One way to test if demand data falsify utility maximization is to see if one can find the numbers in (G).

Afriat's Theorem

Theorem (Afriat)

Given a finite set of demand data $\{(x^j, p^j, w^j)\}_{k=1}^N$, *the following are equivalent:*

1. *There exists a locally nonsatiated utility function which rationalizes the data.*
2. *The data satisfy the Generalized Axiom of Revealed Preference.*
3. *There exist numbers* $v^j, \lambda^j > 0$ *such that*
$$v^k \leq v^j + \lambda^j[p^j \cdot x^k - w^j], \qquad \text{for all } j, k = 1, \ldots, N \tag{G}$$
4. *There exists a continuous, strictly increasing, and concave utility function which rationalizes the data.*

- For finite data, the only implication of preference maximization is GARP; conversely, if the data violate GARP, the consumer is not maximizing any continuous, strictly monotone, and convex preference relation.
- Using a finite number of observations, one cannot distinguish a continuous, strictly monotone, and convex preference from a locally nonsatiated one.
- One way to test if demand data falsify utility maximization is to see if one can find the numbers in (G).
  - These numbers help us construct a function that rationalizes data: each observation has "utils" v^i and a "budget violating penalty/prize" λ^i; every choice is rational given these.

The Afriat Numbers

$\exists v^j, \lambda^j > 0$ *s.t.* $v^k \leq v^j + \lambda^j [p^j \cdot x^k - w^j]$ *for all* $j, k = 1, \ldots, N$ *(G)*

- One uses the numbers v^k and λ^k to construct a function that rationalizes the data;
  - the theorem then says that every choice is rational.

Where are the Afriat's numbers coming from?

The Afriat Numbers

$\exists v^j, \lambda^j > 0$ *s.t.* $v^k \leq v^j + \lambda^j [p^j \cdot x^k - w^j]$ *for all* $j, k = 1, \ldots, N$ *(G)*

- One uses the numbers v^k and λ^k to construct a function that rationalizes the data;
  - the theorem then says that every choice is rational.

Where are the Afriat's numbers coming from?

- Consider utility maximization subject to a budget constraint:

$$x^* \in \arg\max_{x \in \mathbf{R}^n_+} u(x) \text{ s.t. } p \cdot x \leq w.$$

The Afriat Numbers

$\exists v^j, \lambda^j > 0$ *s.t.* $v^k \leq v^j + \lambda^j [p^j \cdot x^k - w^j]$ *for all* $j, k = 1, \ldots, N$ *(G)*

- One uses the numbers v^k and λ^k to construct a function that rationalizes the data;
  - the theorem then says that every choice is rational.

Where are the Afriat's numbers coming from?

- Consider utility maximization subject to a budget constraint:

$$x^* \in \arg\max_{x \in \mathbf{R}^n_+} u(x) \text{ s.t. } p \cdot x \leq w.$$

- The solution maximizes the Lagrangean:

$$L(x, \lambda) = u(x) + \lambda(w - p \cdot x).$$

The Afriat Numbers

$\exists v^j, \lambda^j > 0$ *s.t.* $v^k \leq v^j + \lambda^j [p^j \cdot x^k - w^j]$ *for all* $j, k = 1, \ldots, N$ *(G)*

- One uses the numbers v^k and λ^k to construct a function that rationalizes the data;
  - the theorem then says that every choice is rational.

Where are the Afriat's numbers coming from?

- Consider utility maximization subject to a budget constraint:

$$x^* \in \arg\max_{x \in \mathbf{R}^n_+} u(x) \text{ s.t. } p \cdot x \leq w.$$

- The solution maximizes the Lagrangean:

$$L(x, \lambda) = u(x) + \lambda(w - p \cdot x).$$

  - Let x^* and λ^* be the optimal consumption and multiplier values, respectively.

The Afriat Numbers

$\exists v^j, \lambda^j > 0$ *s.t.* $v^k \leq v^j + \lambda^j [p^j \cdot x^k - w^j]$ *for all* $j, k = 1, \ldots, N$ *(G)*

- One uses the numbers v^k and λ^k to construct a function that rationalizes the data;
  - the theorem then says that every choice is rational.

Where are the Afriat's numbers coming from?

- Consider utility maximization subject to a budget constraint:

$$x^* \in \arg\max_{x \in \mathbf{R}^n_+} u(x) \text{ s.t. } p \cdot x \leq w.$$

- The solution maximizes the Lagrangean:

$$L(x, \lambda) = u(x) + \lambda(w - p \cdot x).$$

  - Let x^* and λ^* be the optimal consumption and multiplier values, respectively.
  - Then we have $L(x^*, \lambda^*) \geq L(x, \lambda^*)$ for all $x \in \mathbf{R}^n_+$. That is,

$$u(x^*) + \lambda^* \underbrace{(w - p \cdot x^*)}_{=0 \text{ since budget constraint binds}} \geq u(x) + \lambda^*(w - p \cdot x).$$

The Afriat Numbers

$\exists v^j, \lambda^j > 0$ *s.t.* $v^k \leq v^j + \lambda^j [p^j \cdot x^k - w^j]$ *for all* $j, k = 1, \ldots, N$ *(G)*

- One uses the numbers v^k and λ^k to construct a function that rationalizes the data;
  - the theorem then says that every choice is rational.

Where are the Afriat's numbers coming from?

- Consider utility maximization subject to a budget constraint:

$$x^* \in \arg\max_{x \in \mathbf{R}^n_+} u(x) \text{ s.t. } p \cdot x \leq w.$$

- The solution maximizes the Lagrangean:

$$L(x, \lambda) = u(x) + \lambda(w - p \cdot x).$$

  - Let x^* and λ^* be the optimal consumption and multiplier values, respectively.
  - Then we have $L(x^*, \lambda^*) \geq L(x, \lambda^*)$ for all $x \in \mathbf{R}^n_+$. That is,

$$u(x^*) + \lambda^* \underbrace{(w - p \cdot x^*)}_{=0 \text{ since budget constraint binds}} \geq u(x) + \lambda^*(w - p \cdot x).$$

  - Thus for all $x \in \mathbf{R}^n_+$

$$u(x) \leq u(x^*) + \lambda^*(p \cdot x - w)$$

Intuition Behind the Afriat Numbers

$\exists v^j, \lambda^j > 0$ *s.t.* $v^k \leq v^j + \lambda^j[p^j \cdot x^k - w^j]$ *for all* $j, k = 1, \ldots, N$ *(G)*

Another intuition behind Afriat's numbers

- Rewrite (G) as
$$v^k - v^j \leq \lambda^j[p^j \cdot x^k - w^j].$$
- Suppose that $p^j \cdot x^k - w^j > 0$. Then x^k was unaffordable at p^j and w^j.
  - Even if x^k yields more "utils" than x^j, at p^j and w^j the penalty for violating the budget constraint (given by $\lambda^j[p^j \cdot x^k - w^j]$) is too large for the change in the value of the choice (given by $v^k - v^j$) to compensate for it.
  - Thus, the decision maker chose x^j over all other alternatives at p^j and w^j.
- Suppose that $p^j \cdot x^k - w^j \leq 0$. Then x^k was affordable at p^j and w^j.
  - But then (G) implies that
$$v^j - v^k \geq \lambda^j[w^j - p^j \cdot x^k].$$
  - So the decision maker chose x^j over x^k because the difference in "utils" is higher than the bonus from having some extra money left over by choosing x^k instead of x^j.

Producers and Production Sets

Producers are profit maximizing firms that buy inputs and use them to produce and then sell outputs.

- The plural is important because most firms produce more than one good.

Producers and Production Sets

Producers are profit maximizing firms that buy inputs and use them to produce and then sell outputs.

- The plural is important because most firms produce more than one good.
- The standard undergraduate textbook description focuses on one ouput and a few inputs (two in most cases).

Producers and Production Sets

Producers are profit maximizing firms that buy inputs and use them to produce and then sell outputs.

- The plural is important because most firms produce more than one good.
- The standard undergraduate textbook description focuses on one ouput and a few inputs (two in most cases).
- In that framework, production is described by a function that takes inputs as the domain and output(s) as the range.

Producers and Production Sets

Producers are profit maximizing firms that buy inputs and use them to produce and then sell outputs.

- The plural is important because most firms produce more than one good.
- The standard undergraduate textbook description focuses on one ouput and a few inputs (two in most cases).
- In that framework, production is described by a function that takes inputs as the domain and output(s) as the range.
- Here we focus on a more general and abstract version of production.

Producers and Production Sets

Producers are profit maximizing firms that buy inputs and use them to produce and then sell outputs.

- The plural is important because most firms produce more than one good.
- The standard undergraduate textbook description focuses on one ouput and a few inputs (two in most cases).
- In that framework, production is described by a function that takes inputs as the domain and output(s) as the range.
- Here we focus on a more general and abstract version of production.

Definition

A production set is a subset $Y \subseteq \mathbf{R}^n$.

Producers and Production Sets

Producers are profit maximizing firms that buy inputs and use them to produce and then sell outputs.

- The plural is important because most firms produce more than one good.
- The standard undergraduate textbook description focuses on one ouput and a few inputs (two in most cases).
- In that framework, production is described by a function that takes inputs as the domain and output(s) as the range.
- Here we focus on a more general and abstract version of production.

Definition

A production set is a subset $Y \subseteq \mathbf{R}^n$.

- $y = (y_1, ..., y_N)$ denotes production (input-output) vectors.

Producers and Production Sets

Producers are profit maximizing firms that buy inputs and use them to produce and then sell outputs.

- The plural is important because most firms produce more than one good.

- The standard undergraduate textbook description focuses on one ouput and a few inputs (two in most cases).
- In that framework, production is described by a function that takes inputs as the domain and output(s) as the range.
- Here we focus on a more general and abstract version of production.

Definition

A production set is a subset $Y \subseteq \mathbf{R}^n$.

- $y = (y_1, ..., y_N)$ denotes production (input-output) vectors.
- A production vector has outputs as non-negative numbers and inputs as non-positive numbers: $y_i \leq 0$ when i is an input, and $y_i \geq 0$ if i is an output.

Producers and Production Sets

Producers are profit maximizing firms that buy inputs and use them to produce and then sell outputs.

- The plural is important because most firms produce more than one good.

- The standard undergraduate textbook description focuses on one ouput and a few inputs (two in most cases).
- In that framework, production is described by a function that takes inputs as the domain and output(s) as the range.
- Here we focus on a more general and abstract version of production.

Definition

A production set is a subset $Y \subseteq \mathbf{R}^n$.

- $y = (y_1, ..., y_N)$ denotes production (input-output) vectors.
- A production vector has outputs as non-negative numbers and inputs as non-positive numbers: $y_i \leq 0$ when i is an input, and $y_i \geq 0$ if i is an output.
- What is $p \cdot y$ in this notation?

Production Set Properties

Definition

$Y \subseteq \mathbf{R}^n$ satisfies:

Draw Pictures.

Production Set Properties

Definition

$Y \subseteq \mathbf{R}^n$ satisfies:

- no free lunch if $Y \cap \mathbf{R}^n_+ \subseteq \{\mathbf{0}_n\}$;

Draw Pictures.

Production Set Properties

Definition

$Y \subseteq \mathbf{R}^n$ satisfies:

- no free lunch if $Y \cap \mathbf{R}^n_+ \subseteq \{\mathbf{0}_n\}$;
- possibility of inaction if $\mathbf{0}_n \in Y$;

Draw Pictures.

Production Set Properties

Definition

$Y \subseteq \mathbf{R}^n$ satisfies:

- no free lunch if $Y \cap \mathbf{R}^n_+ \subseteq \{\mathbf{0}_n\}$;
- possibility of inaction if $\mathbf{0}_n \in Y$;
- free disposal if $y \in Y$ implies $y' \in Y$ for all $y' \leq y$;

Draw Pictures.

Production Set Properties

Definition

$Y \subseteq \mathbf{R}^n$ satisfies:

- no free lunch if $Y \cap \mathbf{R}^n_+ \subseteq \{\mathbf{0}_n\}$;
- possibility of inaction if $\mathbf{0}_n \in Y$;
- free disposal if $y \in Y$ implies $y' \in Y$ for all $y' \leq y$;
- irreversibility if $y \in Y$ and $y \neq \mathbf{0}_n$ imply $-y \notin Y$;

Draw Pictures.

Production Set Properties

Definition

$Y \subseteq \mathbf{R}^n$ satisfies:

- no free lunch if $Y \cap \mathbf{R}^n_+ \subseteq \{\mathbf{0}_n\}$;
- possibility of inaction if $\mathbf{0}_n \in Y$;
- free disposal if $y \in Y$ implies $y' \in Y$ for all $y' \leq y$;
- irreversibility if $y \in Y$ and $y \neq \mathbf{0}_n$ imply $-y \notin Y$;
- nonincreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \in [0, 1]$;

Draw Pictures.

Production Set Properties

Definition

$Y \subseteq \mathbf{R}^n$ satisfies:

- no free lunch if $Y \cap \mathbf{R}^n_+ \subseteq \{\mathbf{0}_n\}$;
- possibility of inaction if $\mathbf{0}_n \in Y$;
- free disposal if $y \in Y$ implies $y' \in Y$ for all $y' \leq y$;
- irreversibility if $y \in Y$ and $y \neq \mathbf{0}_n$ imply $-y \notin Y$;
- nonincreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \in [0, 1]$;
- nondecreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \geq 1$;

Draw Pictures.

Production Set Properties

Definition

$Y \subseteq \mathbf{R}^n$ satisfies:

- no free lunch if $Y \cap \mathbf{R}^n_+ \subseteq \{\mathbf{0}_n\}$;
- possibility of inaction if $\mathbf{0}_n \in Y$;
- free disposal if $y \in Y$ implies $y' \in Y$ for all $y' \leq y$;
- irreversibility if $y \in Y$ and $y \neq \mathbf{0}_n$ imply $-y \notin Y$;
- nonincreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \in [0, 1]$;
- nondecreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \geq 1$;
- constant returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \geq 0$;

Draw Pictures.

Production Set Properties

Definition

$Y \subseteq \mathbf{R}^n$ satisfies:

- no free lunch if $Y \cap \mathbf{R}^n_+ \subseteq \{\mathbf{0}_n\}$;
- possibility of inaction if $\mathbf{0}_n \in Y$;
- free disposal if $y \in Y$ implies $y' \in Y$ for all $y' \leq y$;
- irreversibility if $y \in Y$ and $y \neq \mathbf{0}_n$ imply $-y \notin Y$;
- nonincreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \in [0, 1]$;
- nondecreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \geq 1$;
- constant returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \geq 0$;
- additivity if $y, y' \in Y$ imply $y + y' \in Y$;

Draw Pictures.

Production Set Properties

Definition

$Y \subseteq \mathbf{R}^n$ satisfies:

- no free lunch if $Y \cap \mathbf{R}^n_+ \subseteq \{\mathbf{0}_n\}$;
- possibility of inaction if $\mathbf{0}_n \in Y$;
- free disposal if $y \in Y$ implies $y' \in Y$ for all $y' \leq y$;
- irreversibility if $y \in Y$ and $y \neq \mathbf{0}_n$ imply $-y \notin Y$;
- nonincreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \in [0, 1]$;
- nondecreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \geq 1$;
- constant returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \geq 0$;
- additivity if $y, y' \in Y$ imply $y + y' \in Y$;
- convexity if Y is convex;

Draw Pictures.

Production Set Properties

Definition

$Y \subseteq \mathbf{R}^n$ satisfies:

- no free lunch if $Y \cap \mathbf{R}^n_+ \subseteq \{\mathbf{0}_n\}$;
- possibility of inaction if $\mathbf{0}_n \in Y$;
- free disposal if $y \in Y$ implies $y' \in Y$ for all $y' \leq y$;
- irreversibility if $y \in Y$ and $y \neq \mathbf{0}_n$ imply $-y \notin Y$;
- nonincreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \in [0, 1]$;
- nondecreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \geq 1$;
- constant returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \geq 0$;
- additivity if $y, y' \in Y$ imply $y + y' \in Y$;
- convexity if Y is convex;
- Y is a convex cone if for any $y, y' \in Y$ and $\alpha, \beta \geq 0$, $\alpha y + \beta y' \in Y$.

Draw Pictures.

Production Set Properties Are Related

- Some of these properties are related to each other.

Exercise

Y satisfies additivity and nonincreasing returns if and only if it is a convex cone.

Production Set Properties Are Related

- Some of these properties are related to each other.

Exercise

Y satisfies additivity and nonincreasing returns if and only if it is a convex cone.

Exercise

For any convex $Y \subset \mathbf{R}^n$ such that $\mathbf{0}_n \in Y$, there is a convex $Y' \subset \mathbf{R}^{n+1}$ that satisfies constant returns such that $Y = \{y \in \mathbf{R}^n : (y, -1) \in \mathbf{R}^{n+1}\}$.

Production Functions

Let $y \in \mathbf{R}_+^m$ denote outputs while $x \in \mathbf{R}_+^n$ represent inputs; if the two are related by a function $f : \mathbf{R}_+^n \to \mathbf{R}_+^m$, we write $y = f(x)$ to say that y units of outputs are produced using x amount of the inputs.

Production Functions

Let $y \in \mathbf{R}_+^m$ denote outputs while $x \in \mathbf{R}_+^n$ represent inputs; if the two are related by a function $f : \mathbf{R}_+^n \to \mathbf{R}_+^m$, we write $y = f(x)$ to say that y units of outputs are produced using x amount of the inputs.

- When $m = 1$, this is the familiar one output many inputs production function.

Production Functions

Let $y \in \mathbf{R}_+^m$ denote outputs while $x \in \mathbf{R}_+^n$ represent inputs; if the two are related by a function $f : \mathbf{R}_+^n \to \mathbf{R}_+^m$, we write $y = f(x)$ to say that y units of outputs are produced using x amount of the inputs.

- When $m = 1$, this is the familiar one output many inputs production function.
- Production sets and the familiar production function are related.

Production Functions

Let $y \in \mathbf{R}_+^m$ denote outputs while $x \in \mathbf{R}_+^n$ represent inputs; if the two are related by a function $f : \mathbf{R}_+^n \to \mathbf{R}_+^m$, we write $y = f(x)$ to say that y units of outputs are produced using x amount of the inputs.

- When $m = 1$, this is the familiar one output many inputs production function.
- Production sets and the familiar production function are related.

Exercise

Suppose the firm's production set is generated by a production function $f : \mathbf{R}_+^n \to \mathbf{R}_+^m$, where $\mathbf{R}_+^n$ represents its n inputs and $\mathbf{R}_+$ represents its m outputs. Let

$$Y = \{(-x, y) \in \mathbf{R}_-^n \times \mathbf{R}_+^m : y \leq f(x)\}.$$

Prove the following:

1. Y satisfies no free lunch, possibility of inaction, free disposal, and irreversibility.
2. Suppose $m = 1$. Y satisfies constant returns to scale if and only if f is homogeneous of degree one, i.e. $f(\alpha x) = \alpha f(x)$ for all $\alpha \geq 0$.
3. Suppose $m = 1$. Y satisfies convexity if and only if f is concave.

Transformation Function

- We can describe production sets using a function.

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the transformation function $F : Y \to \mathbf{R}$ is defined by

$$Y = \{y \in Y : F(y) \leq 0 \text{ and } F(y) = 0 \text{ if and only if } y \text{ is on the boundary of } Y\};$$

the transformation frontier is

$$\{y \in \mathbf{R}^n : F(y) = 0\}$$

Transformation Function

- We can describe production sets using a function.

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the transformation function $F : Y \to \mathbf{R}$ is defined by

$$Y = \{y \in Y : F(y) \leq 0 \text{ and } F(y) = 0 \text{ if and only if } y \text{ is on the boundary of } Y\};$$

the transformation frontier is

$$\{y \in \mathbf{R}^n : F(y) = 0\}$$

Definition

Given a differentiable transformation function F and a point on its transformation frontier y, the marginal rate of transformation for goods i and j is given by

$$MRT_{i,j} = \frac{\frac{\partial F(y)}{\partial y_i}}{\frac{\partial F(y)}{\partial y_j}}$$

Transformation Function

- We can describe production sets using a function.

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the transformation function $F : Y \to \mathbf{R}$ is defined by

$$Y = \{y \in Y : F(y) \leq 0 \text{ and } F(y) = 0 \text{ if and only if } y \text{ is on the boundary of } Y\};$$

the transformation frontier is

$$\{y \in \mathbf{R}^n : F(y) = 0\}$$

Definition

Given a differentiable transformation function F and a point on its transformation frontier y, the marginal rate of transformation for goods i and j is given by

$$MRT_{i,j} = \frac{\frac{\partial F(y)}{\partial y_i}}{\frac{\partial F(y)}{\partial y_j}}$$

- Since $F(y) = 0$ we have

$$\frac{\partial F(y)}{\partial y_i} dy_i + \frac{\partial F(y)}{\partial y_j} dy_j = 0$$

Transformation Function

- We can describe production sets using a function.

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the transformation function $F : Y \to \mathbf{R}$ is defined by

$$Y = \{y \in Y : F(y) \leq 0 \text{ and } F(y) = 0 \text{ if and only if } y \text{ is on the boundary of } Y\};$$

the transformation frontier is

$$\{y \in \mathbf{R}^n : F(y) = 0\}$$

Definition

Given a differentiable transformation function F and a point on its transformation frontier y, the marginal rate of transformation for goods i and j is given by

$$MRT_{i,j} = \frac{\frac{\partial F(y)}{\partial y_i}}{\frac{\partial F(y)}{\partial y_j}}$$

- Since $F(y) = 0$ we have

$$\frac{\partial F(y)}{\partial y_i} dy_i + \frac{\partial F(y)}{\partial y_j} dy_j = 0$$

  - MRT is the slope of the transformation frontier at y.

Profit Maximization

Profit Maximizing Assumption

The firm's objective is to choose a production vector on the trasformation frontier as to maximize profits given prices $p \in \mathbf{R}^n_{++}$:

$$\max_{y \in Y} p \cdot y$$

or equivalently

$$\max p \cdot y \qquad \text{subject to} \qquad F(y) \leq 0$$

Profit Maximization

Profit Maximizing Assumption

The firm's objective is to choose a production vector on the trasformation frontier as to maximize profits given prices $p \in \mathbf{R}^n_{++}$:

$$\max_{y \in Y} p \cdot y$$

or equivalently

$$\max p \cdot y \qquad \text{subject to} \qquad F(y) \leq 0$$

- Does this distinguish between revenues and costs? How?

Profit Maximization

Profit Maximizing Assumption

The firm's objective is to choose a production vector on the trasformation frontier as to maximize profits given prices $p \in \mathbf{R}^n_{++}$:

$$\max_{y \in Y} p \cdot y$$

or equivalently

$$\max p \cdot y \qquad \text{subject to} \qquad F(y) \leq 0$$

- Does this distinguish between revenues and costs? How?
- Using the single output production function:

$$\max_{x \geq 0} pf(x) - w \cdot x$$

Profit Maximization

Profit Maximizing Assumption

The firm's objective is to choose a production vector on the trasformation frontier as to maximize profits given prices $p \in \mathbf{R}^n_{++}$:

$$\max_{y \in Y} p \cdot y$$

or equivalently

$$\max p \cdot y \qquad \text{subject to} \qquad F(y) \leq 0$$

- Does this distinguish between revenues and costs? How?
- Using the single output production function:

$$\max_{x \geq 0} pf(x) - w \cdot x$$

  - here $p \in \mathbf{R}_{++}$ is the price of output and $w \in \mathbf{R}^l_{++}$ is the price of inputs.

First Order Conditions For Profit Maximization

$$\max p \cdot y \qquad \textit{subject to} \qquad F(y) = 0$$

Profit Maximizing

First Order Conditions For Profit Maximization

$$\max p \cdot y \qquad \textit{subject to} \qquad F(y) = 0$$

Profit Maximizing

- The FOC are

$$p_i = \lambda \frac{\partial F(y)}{\partial y_i} \text{ for each } i \qquad \text{or} \qquad \underbrace{p}_{1\times n} = \lambda \underbrace{\nabla F(y)}_{1\times n} \text{ in matrix form}$$

First Order Conditions For Profit Maximization

$$\max p \cdot y \qquad \textit{subject to} \qquad F(y) = 0$$

Profit Maximizing

- The FOC are

$$p_i = \lambda \frac{\partial F(y)}{\partial y_i} \text{ for each } i \qquad \text{or} \qquad \underbrace{p}_{1\times n} = \lambda \underbrace{\nabla F(y)}_{1\times n} \text{ in matrix form}$$

- Therefore

$$\frac{1}{\lambda} = \frac{\frac{\partial F(y)}{\partial y_i}}{p_i} \text{ for each } i$$

First Order Conditions For Profit Maximization

$$\max p \cdot y \qquad \textit{subject to} \qquad F(y) = 0$$

Profit Maximizing

- The FOC are

$$p_i = \lambda \frac{\partial F(y)}{\partial y_i} \text{ for each } i \qquad \text{or} \qquad \underbrace{p}_{1\times n} = \lambda \underbrace{\nabla F(y)}_{1\times n} \text{ in matrix form}$$

- Therefore

$$\frac{1}{\lambda} = \frac{\frac{\partial F(y)}{\partial y_i}}{p_i} \text{ for each } i$$

  - the marginal product per dollar spent or received is equal across all goods.

First Order Conditions For Profit Maximization

$$\max p \cdot y \qquad \textit{subject to} \qquad F(y) = 0$$

Profit Maximizing

- The FOC are

$$p_i = \lambda \frac{\partial F(y)}{\partial y_i} \text{ for each } i \qquad \text{or} \qquad \underbrace{p}_{1\times n} = \lambda \underbrace{\nabla F(y)}_{1\times n} \text{ in matrix form}$$

- Therefore

$$\frac{1}{\lambda} = \frac{\frac{\partial F(y)}{\partial y_i}}{p_i} \text{ for each } i$$

  - the marginal product per dollar spent or received is equal across all goods.

- Using this formula for i and j:

$$\frac{\frac{\partial F(y)}{\partial y_i}}{\frac{\partial F(y)}{\partial y_j}} = MRT_{i,j} = \frac{p_i}{p_j} \text{ for each } i, j$$

First Order Conditions For Profit Maximization

$$\max p \cdot y \qquad \textit{subject to} \qquad F(y) = 0$$

Profit Maximizing

- The FOC are

$$p_i = \lambda \frac{\partial F(y)}{\partial y_i} \text{ for each } i \qquad \text{or} \qquad \underbrace{p}_{1\times n} = \lambda \underbrace{\nabla F(y)}_{1\times n} \text{ in matrix form}$$

- Therefore

$$\frac{1}{\lambda} = \frac{\frac{\partial F(y)}{\partial y_i}}{p_i} \text{ for each } i$$

  - the marginal product per dollar spent or received is equal across all goods.

- Using this formula for i and j:

$$\frac{\frac{\partial F(y)}{\partial y_i}}{\frac{\partial F(y)}{\partial y_j}} = MRT_{i,j} = \frac{p_i}{p_j} \text{ for each } i, j$$

  - the Marginal Rate of Transformation equals the price ratio.

Supply Correspondence and Profit Functions

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the supply correspondence $y^* : \mathbf{R}^n_{++} \to \mathbf{R}^n$ is:

$$y^*(p) = \arg\max_{y \in Y} p \cdot y.$$

Supply Correspondence and Profit Functions

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the supply correspondence $y^* : \mathbf{R}^n_{++} \to \mathbf{R}^n$ is:

$$y^*(p) = \arg\max_{y \in Y} p \cdot y.$$

- Tracks the optimal choice as prices change (similar to Walrasian demand).

Supply Correspondence and Profit Functions

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the supply correspondence $y^* : \mathbf{R}^n_{++} \to \mathbf{R}^n$ is:

$$y^*(p) = \arg\max_{y \in Y} p \cdot y.$$

- Tracks the optimal choice as prices change (similar to Walrasian demand).

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the profit function $\pi : \mathbf{R}^n_{++} \to \mathbf{R}$ is:

$$\pi(p) = \max_{y \in Y} p \cdot y.$$

Supply Correspondence and Profit Functions

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the supply correspondence $y^* : \mathbf{R}^n_{++} \to \mathbf{R}^n$ is:

$$y^*(p) = \arg\max_{y \in Y} p \cdot y.$$

- Tracks the optimal choice as prices change (similar to Walrasian demand).

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the profit function $\pi : \mathbf{R}^n_{++} \to \mathbf{R}$ is:

$$\pi(p) = \max_{y \in Y} p \cdot y.$$

- Tracks maximized profits as prices change (similar to indirect utility function).

Supply Correspondence and Profit Functions

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the supply correspondence $y^* : \mathbf{R}^n_{++} \to \mathbf{R}^n$ is:

$$y^*(p) = \arg\max_{y \in Y} p \cdot y.$$

- Tracks the optimal choice as prices change (similar to Walrasian demand).

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the profit function $\pi : \mathbf{R}^n_{++} \to \mathbf{R}$ is:

$$\pi(p) = \max_{y \in Y} p \cdot y.$$

- Tracks maximized profits as prices change (similar to indirect utility function).

Proposition

If Y satisfies non decreasing returns to scale either $\pi(p) \leq 0$ or $\pi(p) = +\infty$.

Proof.

Question 6, Problem Set 5. □

Properties of Supply and Profit Functions

Proposition

Suppose Y is closed and satisfies free disposal. Then:

Properties of Supply and Profit Functions

Proposition

Suppose Y is closed and satisfies free disposal. Then:

- $\pi(\alpha p) = \alpha\pi(p)$ *for all* $\alpha > 0$;

Properties of Supply and Profit Functions

Proposition

Suppose Y is closed and satisfies free disposal. Then:

- $\pi(\alpha p) = \alpha\pi(p)$ *for all* $\alpha > 0$;
- π *is convex in* p;

Properties of Supply and Profit Functions

Proposition

Suppose Y is closed and satisfies free disposal. Then:

- $\pi(\alpha p) = \alpha\pi(p)$ *for all* $\alpha > 0$;
- π *is convex in* p;
- $y^*(\alpha p) = y^*(p)$ *for all* $\alpha > 0$;

Properties of Supply and Profit Functions

Proposition

Suppose Y is closed and satisfies free disposal. Then:

- $\pi(\alpha p) = \alpha\pi(p)$ *for all* $\alpha > 0$;
- π *is convex in* p;
- $y^*(\alpha p) = y^*(p)$ *for all* $\alpha > 0$;
- *if* Y *is convex, then* $y^*(p)$ *is convex;*

Properties of Supply and Profit Functions

Proposition

Suppose Y is closed and satisfies free disposal. Then:

- $\pi(\alpha p) = \alpha\pi(p)$ *for all* $\alpha > 0$;
- π *is convex in* p;
- $y^*(\alpha p) = y^*(p)$ *for all* $\alpha > 0$;
- *if* Y *is convex, then* $y^*(p)$ *is convex;*
- *if* $|y^*(p)| = 1$, *then* π *is differentiable at* p *and* $\nabla\pi(p) = y^*(p)$ *(Hotelling's Lemma).*

Properties of Supply and Profit Functions

Proposition

Suppose Y is closed and satisfies free disposal. Then:

- $\pi(\alpha p) = \alpha\pi(p)$ *for all* $\alpha > 0$;
- π *is convex in* p;
- $y^*(\alpha p) = y^*(p)$ *for all* $\alpha > 0$;
- *if* Y *is convex, then* $y^*(p)$ *is convex;*
- *if* $|y^*(p)| = 1$, *then* π *is differentiable at* p *and* $\nabla\pi(p) = y^*(p)$ *(Hotelling's Lemma).*
- *if* $y^*(p)$ *is differentiable at* p, *then* $Dy^*(p) = D^2\pi(p)$ *is symmetric and positive semidefinite with* $Dy^*(p)p = 0$.

The Profit Function Is Convex

Proof.

Let $p,p' \in \mathbf{R}^n_{++}$ and let the corresponding profit maximizing solutions be y and y'.

The Profit Function Is Convex

Proof.

Let $p, p' \in \mathbf{R}^n_{++}$ and let the corresponding profit maximizing solutions be y and y'.

- For any $\lambda \in (0,1)$ let $\bar{p} = \lambda p + (1-\lambda)\, p'$ and let $\bar{y}$ be the profit maximizing output vector when prices are $\bar{p}$.

The Profit Function Is Convex

Proof.

Let $p, p' \in \mathbf{R}^n_{++}$ and let the corresponding profit maximizing solutions be y and y'.

- For any $\lambda \in (0,1)$ let $\bar{p} = \lambda p + (1-\lambda)\, p'$ and let $\bar{y}$ be the profit maximizing output vector when prices are $\bar{p}$.
- By "revealed preferences"

$$p \cdot y \geq p \cdot \bar{y} \qquad \text{and} \qquad p' \cdot y' \geq p' \cdot \bar{y}$$

The Profit Function Is Convex

Proof.

Let $p, p' \in \mathbf{R}^n_{++}$ and let the corresponding profit maximizing solutions be y and y'.

- For any $\lambda \in (0,1)$ let $\bar{p} = \lambda p + (1-\lambda)\, p'$ and let $\bar{y}$ be the profit maximizing output vector when prices are $\bar{p}$.
- By "revealed preferences"

$$p \cdot y \geq p \cdot \bar{y} \qquad \text{and} \qquad p' \cdot y' \geq p' \cdot \bar{y}$$

  - why?

The Profit Function Is Convex

Proof.

Let $p, p' \in \mathbf{R}^n_{++}$ and let the corresponding profit maximizing solutions be y and y'.

- For any $\lambda \in (0,1)$ let $\bar{p} = \lambda p + (1-\lambda)\, p'$ and let $\bar{y}$ be the profit maximizing output vector when prices are $\bar{p}$.
- By "revealed preferences"

$$p \cdot y \geq p \cdot \bar{y} \qquad \text{and} \qquad p' \cdot y' \geq p' \cdot \bar{y}$$

  - why?
- multiply these inequalities by λ and $1-\lambda$

$$\lambda p \cdot y \geq \lambda p \cdot \bar{y} \qquad \text{and} \qquad (1-\lambda)\, p' \cdot y' \geq (1-\lambda)\, p' \cdot \bar{y}$$

The Profit Function Is Convex

Proof.

Let $p, p' \in \mathbf{R}^n_{++}$ and let the corresponding profit maximizing solutions be y and y'.

- For any $\lambda \in (0,1)$ let $\bar{p} = \lambda p + (1-\lambda)\, p'$ and let $\bar{y}$ be the profit maximizing output vector when prices are $\bar{p}$.
- By "revealed preferences"

$$p \cdot y \geq p \cdot \bar{y} \qquad \text{and} \qquad p' \cdot y' \geq p' \cdot \bar{y}$$

  - why?
- multiply these inequalities by λ and $1-\lambda$

$$\lambda p \cdot y \geq \lambda p \cdot \bar{y} \qquad \text{and} \qquad (1-\lambda)\, p' \cdot y' \geq (1-\lambda)\, p' \cdot \bar{y}$$

- summing up

$$\lambda p \cdot y + (1-\lambda)\, p' \cdot y' \geq [\lambda p + (1-\lambda)\, p'] \cdot \bar{y}$$

The Profit Function Is Convex

Proof.

Let $p, p' \in \mathbf{R}^n_{++}$ and let the corresponding profit maximizing solutions be y and y'.

- For any $\lambda \in (0,1)$ let $\bar{p} = \lambda p + (1-\lambda) p'$ and let $\bar{y}$ be the profit maximizing output vector when prices are $\bar{p}$.
- By "revealed preferences"

$$p \cdot y \geq p \cdot \bar{y} \qquad \text{and} \qquad p' \cdot y' \geq p' \cdot \bar{y}$$

  - why?
- multiply these inequalities by λ and $1-\lambda$

$$\lambda p \cdot y \geq \lambda p \cdot \bar{y} \qquad \text{and} \qquad (1-\lambda) p' \cdot y' \geq (1-\lambda) p' \cdot \bar{y}$$

- summing up

$$\lambda p \cdot y + (1-\lambda) p' \cdot y' \geq [\lambda p + (1-\lambda) p'] \cdot \bar{y}$$

- using the definition of profit function:

$$\lambda \pi(p) + (1-\lambda) \pi(p') \geq \pi(\lambda p + (1-\lambda) p')$$

proving convexity of $\pi(p)$. □

The Supply Correspondence Is Convex

Proof.

Let $p \in \mathbf{R}^n_{++}$ and let $y, y' \in y^*(p)$.

The Supply Correspondence Is Convex

Proof.

Let $p \in \mathbf{R}^n_{++}$ and let $y, y' \in y^*(p)$.
We need to show that if Y is convex then

$$\lambda y + (1-\lambda)\, y' \in y^*(p) \qquad \text{for any } \lambda \in (0,1)$$

The Supply Correspondence Is Convex

Proof.

Let $p \in \mathbf{R}^n_{++}$ and let $y, y' \in y^*(p)$.
We need to show that if Y is convex then

$$\lambda y + (1-\lambda)\, y' \in y^*(p) \qquad \text{for any } \lambda \in (0,1)$$

- By definition:

$$p \cdot y \geq p \cdot \bar{y} \quad \text{for any } \bar{y} \in Y$$

The Supply Correspondence Is Convex

Proof.

Let $p \in \mathbf{R}^n_{++}$ and let $y, y' \in y^*(p)$.
We need to show that if Y is convex then

$$\lambda y + (1-\lambda)\, y' \in y^*(p) \qquad \text{for any } \lambda \in (0,1)$$

- By definition:

$$p \cdot y \geq p \cdot \bar{y} \quad \text{for any } \bar{y} \in Y \qquad \text{and} \qquad p \cdot y' \geq p \cdot \bar{y} \quad \text{for any } \bar{y} \in Y$$

The Supply Correspondence Is Convex

Proof.

Let $p \in \mathbf{R}^n_{++}$ and let $y, y' \in y^*(p)$.
We need to show that if Y is convex then

$$\lambda y + (1-\lambda)\, y' \in y^*(p) \qquad \text{for any } \lambda \in (0,1)$$

- By definition:

$$p \cdot y \geq p \cdot \bar{y} \quad \text{for any } \bar{y} \in Y \qquad \text{and} \qquad p \cdot y' \geq p \cdot \bar{y} \quad \text{for any } \bar{y} \in Y$$

- multiplying by λ and $1-\lambda$ we get

$$\lambda p \cdot y \geq \lambda p \cdot \bar{y} \qquad \text{and} \qquad (1-\lambda)\, p \cdot y' \geq (1-\lambda)\, p \cdot \bar{y}$$

The Supply Correspondence Is Convex

Proof.

Let $p \in \mathbf{R}^n_{++}$ and let $y, y' \in y^*(p)$.
We need to show that if Y is convex then

$$\lambda y + (1-\lambda)\, y' \in y^*(p) \qquad \text{for any } \lambda \in (0,1)$$

- By definition:

$$p \cdot y \geq p \cdot \bar{y} \quad \text{for any } \bar{y} \in Y \qquad \text{and} \qquad p \cdot y' \geq p \cdot \bar{y} \quad \text{for any } \bar{y} \in Y$$

- multiplying by λ and $1-\lambda$ we get

$$\lambda p \cdot y \geq \lambda p \cdot \bar{y} \qquad \text{and} \qquad (1-\lambda)\, p \cdot y' \geq (1-\lambda)\, p \cdot \bar{y}$$

- Therefore, summing up, we have

$$\lambda p \cdot y + (1-\lambda)\, p \cdot y' \geq [\lambda + (1-\lambda)]\, p \cdot \bar{y}$$

The Supply Correspondence Is Convex

Proof.

Let $p \in \mathbf{R}^n_{++}$ and let $y, y' \in y^*(p)$.
We need to show that if Y is convex then

$$\lambda y + (1-\lambda)\, y' \in y^*(p) \qquad \text{for any } \lambda \in (0,1)$$

- By definition:

$$p \cdot y \geq p \cdot \bar{y} \quad \text{for any } \bar{y} \in Y \qquad \text{and} \qquad p \cdot y' \geq p \cdot \bar{y} \quad \text{for any } \bar{y} \in Y$$

- multiplying by λ and $1-\lambda$ we get

$$\lambda p \cdot y \geq \lambda p \cdot \bar{y} \qquad \text{and} \qquad (1-\lambda)\, p \cdot y' \geq (1-\lambda)\, p \cdot \bar{y}$$

- Therefore, summing up, we have

$$\lambda p \cdot y + (1-\lambda)\, p \cdot y' \geq [\lambda + (1-\lambda)]\, p \cdot \bar{y}$$

- Rearranging:

$$p \cdot [\lambda y + (1-\lambda)\, y'] \geq p \cdot \bar{y}$$

proving convexity of $y^*(p)$. □

Hotelling's Lemma

if $|y^*(p)| = 1$, then π is differentiable at p and $\nabla\pi(p) = y^*(p)$

Proof.

Suppose $y^*(p)$ is the unique solution to $\max p \cdot y$ subject to $F(y) = 0$.

Hotelling's Lemma

if $|y^*(p)| = 1$, then π is differentiable at p and $\nabla\pi(p) = y^*(p)$

Proof.

Suppose $y^*(p)$ is the unique solution to $\max p \cdot y$ subject to $F(y) = 0$.

- The Envelope Theorem says

$$D_q\phi(x^*(\bar{q}); \bar{q}) = D_q\phi(x, q)|_{x=x^*(q), q=\bar{q}} - [\lambda^*(\bar{q})]^\top D_q F(x, q)|_{x=x^*(\bar{q}), q=\bar{q}}$$

Hotelling's Lemma

if $|y^*(p)| = 1$, then π is differentiable at p and $\nabla\pi(p) = y^*(p)$

Proof.

Suppose $y^*(p)$ is the unique solution to $\max p \cdot y$ subject to $F(y) = 0$.

- The Envelope Theorem says

$$D_q\phi(x^*(\bar{q}); \bar{q}) = D_q\phi(x, q)|_{x=x^*(q), q=\bar{q}} - [\lambda^*(\bar{q})]^\top D_q F(x, q)|_{x=x^*(\bar{q}), q=\bar{q}}$$

- In our setting,

Hotelling's Lemma

if $|y^*(p)| = 1$, then π is differentiable at p and $\nabla\pi(p) = y^*(p)$

Proof.

Suppose $y^*(p)$ is the unique solution to $\max p \cdot y$ subject to $F(y) = 0$.

- The Envelope Theorem says

$$D_q\phi(x^*(\bar{q}); \bar{q}) = D_q\phi(x, q)|_{x=x^*(q), q=\bar{q}} - [\lambda^*(\bar{q})]^\top D_q F(x, q)|_{x=x^*(\bar{q}), q=\bar{q}}$$

- In our setting,
  - $\phi(x, q) = p \cdot y$,

Hotelling's Lemma

if $|y^*(p)| = 1$, then π is differentiable at p and $\nabla\pi(p) = y^*(p)$

Proof.

Suppose $y^*(p)$ is the unique solution to $\max p \cdot y$ subject to $F(y) = 0$.

- The Envelope Theorem says

$$D_q\phi(x^*(\bar{q}); \bar{q}) = D_q\phi(x, q)|_{x=x^*(q), q=\bar{q}} - [\lambda^*(\bar{q})]^\top D_q F(x, q)|_{x=x^*(\bar{q}), q=\bar{q}}$$

- In our setting,
  - $\phi(x, q) = p \cdot y$,
  - $F(x, q) = F(y)$, and

Hotelling's Lemma

if $|y^*(p)| = 1$, then π is differentiable at p and $\nabla\pi(p) = y^*(p)$

Proof.

Suppose $y^*(p)$ is the unique solution to $\max p \cdot y$ subject to $F(y) = 0$.

- The Envelope Theorem says

$$D_q\phi(x^*(\bar{q}); \bar{q}) = D_q\phi(x, q)|_{x=x^*(q), q=\bar{q}} - [\lambda^*(\bar{q})]^\top D_q F(x, q)|_{x=x^*(\bar{q}), q=\bar{q}}$$

- In our setting,
  - $\phi(x, q) = p \cdot y$,
  - $F(x, q) = F(y)$, and
  - $\phi(x^*(\bar{q}); \bar{q}) = \pi(p)$.

Hotelling's Lemma

if $|y^*(p)| = 1$, then π is differentiable at p and $\nabla\pi(p) = y^*(p)$

Proof.

Suppose $y^*(p)$ is the unique solution to $\max p \cdot y$ subject to $F(y) = 0$.

- The Envelope Theorem says

$$D_q\phi(x^*(\bar{q}); \bar{q}) = D_q\phi(x, q)|_{x=x^*(q), q=\bar{q}} - [\lambda^*(\bar{q})]^\top D_q F(x, q)|_{x=x^*(\bar{q}), q=\bar{q}}$$

- In our setting,
  - $\phi(x, q) = p \cdot y$,
  - $F(x, q) = F(y)$, and
  - $\phi(x^*(\bar{q}); \bar{q}) = \pi(p)$.
- Thus, by the envelope theorem:

$$\nabla\pi(p) = D_p(p \cdot y)|_{y=y^*(p)} - [\lambda^*(p)]^\top D_p F(y)|_{y=y^*(p)}$$

Hotelling's Lemma

if $|y^*(p)| = 1$, then π is differentiable at p and $\nabla\pi(p) = y^*(p)$

Proof.

Suppose $y^*(p)$ is the unique solution to $\max p \cdot y$ subject to $F(y) = 0$.

- The Envelope Theorem says

$$D_q\phi(x^*(\bar{q}); \bar{q}) = D_q\phi(x, q)|_{x=x^*(q), q=\bar{q}} - [\lambda^*(\bar{q})]^\top D_q F(x, q)|_{x=x^*(\bar{q}), q=\bar{q}}$$

- In our setting,
  - $\phi(x, q) = p \cdot y$,
  - $F(x, q) = F(y)$, and
  - $\phi(x^*(\bar{q}); \bar{q}) = \pi(p)$.
- Thus, by the envelope theorem:

$$\nabla\pi(p) = D_p(p \cdot y)|_{y=y^*(p)} - [\lambda^*(p)]^\top D_p F(y)|_{y=y^*(p)} = y|_{y=y^*(p)} - [\lambda^*(p)]^\top 0$$

Hotelling's Lemma

if $|y^*(p)| = 1$, then π is differentiable at p and $\nabla\pi(p) = y^*(p)$

Proof.

Suppose $y^*(p)$ is the unique solution to $\max p \cdot y$ subject to $F(y) = 0$.

- The Envelope Theorem says

$$D_q\phi(x^*(\bar{q}); \bar{q}) = D_q\phi(x, q)|_{x=x^*(q), q=\bar{q}} - [\lambda^*(\bar{q})]^\top D_q F(x, q)|_{x=x^*(\bar{q}), q=\bar{q}}$$

- In our setting,
  - $\phi(x, q) = p \cdot y$,
  - $F(x, q) = F(y)$, and
  - $\phi(x^*(\bar{q}); \bar{q}) = \pi(p)$.
- Thus, by the envelope theorem:

$$\nabla\pi(p) = D_p(p \cdot y)|_{y=y^*(p)} - [\lambda^*(p)]^\top D_p F(y)|_{y=y^*(p)} = y|_{y=y^*(p)} - [\lambda^*(p)]^\top 0$$

  - because $p \cdot y$ is linear in p and $D_p F(y) = 0$.

Hotelling's Lemma

if $|y^*(p)| = 1$, then π is differentiable at p and $\nabla\pi(p) = y^*(p)$

Proof.

Suppose $y^*(p)$ is the unique solution to $\max p \cdot y$ subject to $F(y) = 0$.

- The Envelope Theorem says

$$D_q\phi(x^*(\bar{q}); \bar{q}) = D_q\phi(x, q)|_{x=x^*(q), q=\bar{q}} - [\lambda^*(\bar{q})]^\top D_q F(x, q)|_{x=x^*(\bar{q}), q=\bar{q}}$$

- In our setting,
  - $\phi(x, q) = p \cdot y$,
  - $F(x, q) = F(y)$, and
  - $\phi(x^*(\bar{q}); \bar{q}) = \pi(p)$.
- Thus, by the envelope theorem:

$$\nabla\pi(p) = D_p(p \cdot y)|_{y=y^*(p)} - [\lambda^*(p)]^\top D_p F(y)|_{y=y^*(p)} = y|_{y=y^*(p)} - [\lambda^*(p)]^\top 0$$

  - because $p \cdot y$ is linear in p and $D_p F(y) = 0$.
- Therefore

$$\nabla\pi(p) = y^*(p)$$

as desired. □

Law of Supply

Remark

If $y^*(p)$ is differentiable at p, then $Dy^*(p) = D^2\pi(p)$ is positive semidefinite.

Write the Lagrangian

$$L = p \cdot y - \lambda F(y)$$

Law of Supply

Remark

If $y^*(p)$ is differentiable at p, then $Dy^*(p) = D^2\pi(p)$ is positive semidefinite.

Write the Lagrangian

$$L = p \cdot y - \lambda F(y)$$

- *By the Envelope Theorem:*

$$\frac{\partial \pi(p)}{\partial p_i} = \left.\frac{\partial L}{\partial p_i}\right|_{y=y^*}$$

Law of Supply

Remark

If $y^*(p)$ is differentiable at p, then $Dy^*(p) = D^2\pi(p)$ is positive semidefinite.

Write the Lagrangian

$$L = p \cdot y - \lambda F(y)$$

- *By the Envelope Theorem:*

$$\frac{\partial \pi(p)}{\partial p_i} = \left.\frac{\partial L}{\partial p_i}\right|_{y=y^*} = y_i^*(p)$$

Law of Supply

Remark

If $y^*(p)$ is differentiable at p, then $Dy^*(p) = D^2\pi(p)$ is positive semidefinite.

Write the Lagrangian

$$L = p \cdot y - \lambda F(y)$$

- *By the Envelope Theorem:*

$$\frac{\partial \pi(p)}{\partial p_i} = \left.\frac{\partial L}{\partial p_i}\right|_{y=y^*} = y_i^*(p)$$

- *Therefore, we have*

$$\frac{\partial^2 \pi(p)}{\partial p_i} = \frac{\partial y_i^*(p)}{\partial p_i}$$

Law of Supply

Remark

If $y^*(p)$ is differentiable at p, then $Dy^*(p) = D^2\pi(p)$ is positive semidefinite.

Write the Lagrangian

$$L = p \cdot y - \lambda F(y)$$

- *By the Envelope Theorem:*

$$\frac{\partial \pi(p)}{\partial p_i} = \left.\frac{\partial L}{\partial p_i}\right|_{y=y^*} = y_i^*(p)$$

- *Therefore, we have*

$$\frac{\partial^2 \pi(p)}{\partial p_i} = \frac{\partial y_i^*(p)}{\partial p_i} \geq 0$$

Law of Supply

Remark

If $y^*(p)$ is differentiable at p, then $Dy^*(p) = D^2\pi(p)$ is positive semidefinite.

Write the Lagrangian

$$L = p \cdot y - \lambda F(y)$$

- *By the Envelope Theorem:*

$$\frac{\partial \pi(p)}{\partial p_i} = \left.\frac{\partial L}{\partial p_i}\right|_{y=y^*} = y_i^*(p)$$

- *Therefore, we have*

$$\frac{\partial^2 \pi(p)}{\partial p_i} = \frac{\partial y_i^*(p)}{\partial p_i} \geq 0$$

  - *where the inequality follows from convexity of the profit function.*

Law of Supply

Remark

If $y^*(p)$ is differentiable at p, then $Dy^*(p) = D^2\pi(p)$ is positive semidefinite.

Write the Lagrangian

$$L = p \cdot y - \lambda F(y)$$

- *By the Envelope Theorem:*

$$\frac{\partial \pi(p)}{\partial p_i} = \left.\frac{\partial L}{\partial p_i}\right|_{y=y^*} = y_i^*(p)$$

- *Therefore, we have*

$$\frac{\partial^2 \pi(p)}{\partial p_i} = \frac{\partial y_i^*(p)}{\partial p_i} \geq 0$$

  - *where the inequality follows from convexity of the profit function.*

- This is called the Law of Supply: quantity responds in the same direction as prices.

Law of Supply

Remark

If $y^*(p)$ is differentiable at p, then $Dy^*(p) = D^2\pi(p)$ is positive semidefinite.

Write the Lagrangian

$$L = p \cdot y - \lambda F(y)$$

- *By the Envelope Theorem:*

$$\frac{\partial \pi(p)}{\partial p_i} = \left.\frac{\partial L}{\partial p_i}\right|_{y=y^*} = y_i^*(p)$$

- *Therefore, we have*

$$\frac{\partial^2 \pi(p)}{\partial p_i} = \frac{\partial y_i^*(p)}{\partial p_i} \geq 0$$

  - *where the inequality follows from convexity of the profit function.*

- This is called the Law of Supply: quantity responds in the same direction as prices.
  - Notice that here y_i can be either input or output.

Law of Supply

Remark

If $y^*(p)$ is differentiable at p, then $Dy^*(p) = D^2\pi(p)$ is positive semidefinite.

Write the Lagrangian

$$L = p \cdot y - \lambda F(y)$$

- *By the Envelope Theorem:*

$$\frac{\partial \pi(p)}{\partial p_i} = \left.\frac{\partial L}{\partial p_i}\right|_{y=y^*} = y_i^*(p)$$

- *Therefore, we have*

$$\frac{\partial^2 \pi(p)}{\partial p_i} = \frac{\partial y_i^*(p)}{\partial p_i} \geq 0$$

  - *where the inequality follows from convexity of the profit function.*

- This is called the Law of Supply: quantity responds in the same direction as prices.
  - Notice that here y_i can be either input or output.
    - What does this mean for outputs?

Law of Supply

Remark

If $y^*(p)$ is differentiable at p, then $Dy^*(p) = D^2\pi(p)$ is positive semidefinite.

Write the Lagrangian

$$L = p \cdot y - \lambda F(y)$$

- *By the Envelope Theorem:*

$$\frac{\partial \pi(p)}{\partial p_i} = \left.\frac{\partial L}{\partial p_i}\right|_{y=y^*} = y_i^*(p)$$

- *Therefore, we have*

$$\frac{\partial^2 \pi(p)}{\partial p_i} = \frac{\partial y_i^*(p)}{\partial p_i} \geq 0$$

  - *where the inequality follows from convexity of the profit function.*

- This is called the Law of Supply: quantity responds in the same direction as prices.
  - Notice that here y_i can be either input or output.
    - What does this mean for outputs?
    - What does this mean for inputs?

Factor Demand, Supply, and Profit Function

- The previous concepts can be stated using the production function notation.

Factor Demand, Supply, and Profit Function

- The previous concepts can be stated using the production function notation.

Definition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's factor demand is

$$x^*(p, w) = \arg\max_x \{py - w \cdot x \text{ subject to } f(x) = y\}$$

Factor Demand, Supply, and Profit Function

- The previous concepts can be stated using the production function notation.

Definition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's factor demand is

$$x^*(p, w) = \arg\max_x \{py - w \cdot x \text{ subject to } f(x) = y\} = \arg\max_x pf(x) - w \cdot x.$$

Factor Demand, Supply, and Profit Function

- The previous concepts can be stated using the production function notation.

Definition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's factor demand is

$$x^*(p, w) = \arg\max_x \{py - w \cdot x \text{ subject to } f(x) = y\} = \arg\max_x pf(x) - w \cdot x.$$

Definition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's supply $y^* : \mathbf{R}^n_+ \to \mathbf{R}$ is defined by

$$y^*(p, w) = f(x^*(p, w)).$$

Factor Demand, Supply, and Profit Function

- The previous concepts can be stated using the production function notation.

Definition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's factor demand is

$$x^*(p, w) = \arg\max_x \{py - w \cdot x \text{ subject to } f(x) = y\} = \arg\max_x pf(x) - w \cdot x.$$

Definition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's supply $y^* : \mathbf{R}^n_+ \to \mathbf{R}$ is defined by

$$y^*(p, w) = f(x^*(p, w)).$$

Definition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's profit function $\pi : \mathbf{R}_{++} \times \mathbf{R}^n_{++} \to \mathbf{R}$ is defined by

$$\pi(p, w) = py^*(p, w) - w \cdot x^*(p, w).$$

Factor Demand Properties

- Given these definitions, the following results "translate" the results for output sets to production functions.

Proposition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$,

1. *$\pi(p, w)$ is convex in (p, w).*
2. *$y^*(p, w)$ is non decreasing in p (i.e. $\frac{\partial y^*(p,w)}{\partial p} \geq 0$) and $x^*(p, w)$ is non increasing in w (i.e. $\frac{\partial x_i^*(p,w)}{\partial w_i} \leq 0$) (Hotelling's Lemma).*

Factor Demand Properties

- Given these definitions, the following results "translate" the results for output sets to production functions.

Proposition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$,

1. *$\pi(p, w)$ is convex in (p, w).*
2. *$y^*(p, w)$ is non decreasing in p (i.e. $\frac{\partial y^*(p,w)}{\partial p} \geq 0$) and $x^*(p, w)$ is non increasing in w (i.e. $\frac{\partial x_i^*(p,w)}{\partial w_i} \leq 0$) (Hotelling's Lemma).*

Proof.

Problem 7a,b; Problem Set 4. □

Cost Minimization

Cost Minimizing

- Consider the single output case and suppose the firm wants to deliver a given output quantity at the lowest possible costs. The firm solves

$$\min w \cdot x \qquad \textit{subject to} \qquad f(x) = y$$

Cost Minimization

Cost Minimizing

- Consider the single output case and suppose the firm wants to deliver a given output quantity at the lowest possible costs. The firm solves

$$\min w \cdot x \qquad \textit{subject to} \qquad f(x) = y$$

- This has no simple equivalent in the output vector notation.

Cost Minimization

Cost Minimizing

- Consider the single output case and suppose the firm wants to deliver a given output quantity at the lowest possible costs. The firm solves

$$\min w \cdot x \qquad \textit{subject to} \qquad f(x) = y$$

- This has no simple equivalent in the output vector notation.

Definition

Given $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's conditional factor demand is

$$x^*(w, y) = \arg\min \{w \cdot x \text{ subject to } f(x) = y\};$$

Cost Minimization

Cost Minimizing

- Consider the single output case and suppose the firm wants to deliver a given output quantity at the lowest possible costs. The firm solves

$$\min w \cdot x \qquad \textit{subject to} \qquad f(x) = y$$

- This has no simple equivalent in the output vector notation.

Definition

Given $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's conditional factor demand is

$$x^*(w, y) = \arg\min \{w \cdot x \text{ subject to } f(x) = y\};$$

Definition

Given $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's cost function $C : \mathbf{R}^n_{++} \times \mathbf{R}_+ \to \mathbf{R}$ is defined by

$$C(w, y) = w \cdot x^*(w, y).$$

Properties of Cost Functions

Proposition

Given a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$*, the corresponding cost function* $C(w, y)$ *is concave in* w*.*

Proof.

Question 7c; Problem Set 4. (Hint: use a 'revealed preferences' argument) □

Properties of Cost Functions

Proposition

Given a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$*, the corresponding cost function* $C(w, y)$ *is concave in* w*.*

Proof.

Question 7c; Problem Set 4. (Hint: use a 'revealed preferences' argument) □

Shephard's Lemma

Properties of Cost Functions

Proposition

Given a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, *the corresponding cost function* $C(w, y)$ *is concave in* w.

Proof.

Question 7c; Problem Set 4. (Hint: use a 'revealed preferences' argument) □

Shephard's Lemma

- Write the Lagrangian

$$L = w \cdot x \ - \lambda \left[f\left(x\right) - y \right]$$

Properties of Cost Functions

Proposition

Given a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, *the corresponding cost function* $C(w, y)$ *is concave in* w.

Proof.

Question 7c; Problem Set 4. (Hint: use a 'revealed preferences' argument) □

Shephard's Lemma

- Write the Lagrangian
$$L = w \cdot x \ - \lambda \left[f(x) - y \right]$$
- by the Envelope Theorem
$$\frac{\partial C(w, y)}{\partial w_i} = \frac{\partial L}{\partial w_i} = x_i^* (w, y)$$

Properties of Cost Functions

Proposition

Given a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, *the corresponding cost function* $C(w, y)$ *is concave in* w.

Proof.

Question 7c; Problem Set 4. (Hint: use a 'revealed preferences' argument) □

Shephard's Lemma

- Write the Lagrangian
$$L = w \cdot x \ - \lambda \left[f(x) - y \right]$$
- by the Envelope Theorem
$$\frac{\partial C(w, y)}{\partial w_i} = \frac{\partial L}{\partial w_i} = x_i^* (w, y)$$

Conditional factor demands are downward sloping

Differentiating one more time: $\frac{\partial C^2(w,y)}{\partial w_i \partial w_i} = \frac{\partial x_i^*(w,y)}{\partial w_i} \leq 0$ where the inequality follows concavity of $C(w, y)$.

Next Class

- Decision Making Under Uncertainty
- Expected Utility Representation